GAYATHIRI V

Full Stack Developer | AI-Integrated Applications

+91 9360221678 | gayathiri2k4@gmail.com | LinkedIn | GitHub | Portfolio | Namakkal, Tamil Nadu

PROFESSIONAL SUMMARY

B.Tech Computer Technology graduate (2026) with hands-on experience in full-stack development using Java, React.js, Node.js, and FastAPI, and a strong foundation in integrating machine learning models into production-grade applications. Proven ability to design RESTful APIs, model databases, and ship end-to-end features — from a multimodal emotion recognition system achieving 85% fusion accuracy to deployed web tools with live users. Quick learner with a bias for building real things.

TECHNICAL SKILLS

Languages: Java (primary), JavaScript, Python, C, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend & APIs: Node.js, Express.js, FastAPI, RESTful API Design, Microservices, JWT Authentication
AI / ML: TensorFlow, Keras, OpenCV, LangChain, RAG, HuggingFace, Generative AI
Databases: MongoDB, MySQL, SQLite
DevOps & Deployment: Docker (containerisation), GitHub Actions (CI/CD), Git, Vercel, Render
Tools: VS Code, GitHub, Antigravity

EXPERIENCE

Full Stack Development Intern — **TechnoHacks Solutions Pvt. Ltd.** *Feb 2026 – Mar 2026*
Remote | Full Stack Development

- Built and deployed a full-stack MERN application (Node.js, Express.js, MongoDB, React.js) with JWT-based authentication, role-based access, and a complete RESTful API layer covering GET, POST, PUT, and DELETE operations.
- Designed 6+ API endpoints with CRUD database operations and validated request/response cycles using systematic testing — applying real-world backend security patterns and clean code standards.
- Delivered a responsive UI using Flexbox and CSS Grid layouts, managed full project lifecycle via Git version control, and submitted production-ready documentation within the internship timeline.

PROJECTS

Academic Project

Adaptive Confidence-Based Audio-Visual Emotion Fusion [GitHub] [Live Demo]
Python · TensorFlow · Keras · OpenCV · FastAPI · Streamlit · Docker

- Designed a multimodal emotion recognition system fusing audio and visual data using an adaptive confidence-based weighting mechanism, achieving ~85% weighted fusion accuracy vs. 77% single-modal baseline across 3 multilingual datasets (EMO-DB, EMOVO, SHEMO).
- Built audio emotion recognition with MFCC feature extraction using TensorFlow/Keras deep learning classifiers, enabling multi-language speech emotion classification across 6 emotion classes.
- Developed a video emotion recognition module using OpenCV and CNN-based facial expression analysis, maintaining stable detection under varying lighting, occlusion, and pose variation across 500+ test samples.
- Introduced the Emotion Reliability Index (ERI) — a custom evaluation metric combining accuracy, agreement score, and temporal stability — enabling interpretable and trustworthy multimodal predictions; containerised the full pipeline using Docker for reproducible local deployment.

Personal Projects

AI-Based Student Skill & Mentorship Platform [GitHub]

Java · React.js · Node.js · MongoDB · HuggingFace · Gemini API · GitHub Actions

- Built a full-stack mentorship platform integrating HuggingFace sentence-transformers for NLP-driven skill gap analysis, matching students to mentors based on embedding similarity — reducing manual mentor-matching effort by ~60%.
- Designed and exposed 12+ RESTful API endpoints (authentication, profile management, skill assessment, mentor recommendations), secured with JWT session management and role-based access control for 2 user roles.
- Modelled a normalised MongoDB schema for student profiles, mentor listings, and sessions; applied indexed queries to reduce data retrieval time for high-cardinality searches.
- Implemented CI/CD pipeline using GitHub Actions to automate build and deployment on every push; configured dynamic React dashboards with real-time skill progress tracking and responsive Tailwind CSS layouts.

Metadata Image & Video Anonymizer [GitHub] [Live Demo]

Python · FastAPI · OpenCV · React.js · Docker · Vercel · GitHub Actions

- Developed a FastAPI microservice that detects and strips EXIF, XMP, and embedded metadata from images, videos, and documents, eliminating PII leakage before file sharing — processing 5 file formats (JPEG, PNG, MP4, MOV, PDF).
- Implemented face detection and blurring using OpenCV Haar cascades for optional visual anonymisation, extending the tool into an AI-assisted privacy pipeline with ~95% face detection accuracy.
- Built asynchronous file processing pipelines using Python async I/O; containerised the backend service using Docker for consistent local and production environments.
- Deployed React frontend to Vercel with GitHub Actions CI/CD — automating build, test, and deployment on every commit; features side-by-side original vs. anonymised metadata preview and one-click sanitised file download.

EDUCATION

B.Tech — Computer Technology *CGPA: 8.28 / 10.0 | April 2026*

Bannari Amman Institute of Technology, Sathyamangalam, Tamil Nadu

Relevant Coursework: OOP (Java), Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems, Machine Learning

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Introduction to LangChain – Python | LangChain Academy (Completed)
- Currently pursuing: AWS Cloud Practitioner Essentials (In Progress)

PROFILES & LINKS

Portfolio: gayathiriportfolio-nine.vercel.app **GitHub:** github.com/gayathirivenkatesh

LinkedIn: linkedin.com/in/gayathirivenkatesh